Precambrian Research 170 (2009) 256–266

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Geochronology and paleomagnetism of mafic igneous rocks in the Olenek Uplift, northern Siberia: Implications for Mesoproterozoic supercontinents and paleogeography

Michael T.D. Wingate[a,b,*], Sergei A. Pisarevsky[a,c], Dmitry P. Gladkochub[d], Tatiana V. Donskaya[d], Konstantine M. Konstantinov[e], Anatoly M. Mazukabzov[d], Arkady M. Stanevich[d]

[a] Tectonics Special Research Centre, The University of Western Australia, Crawley, WA 6009, Australia
[b] Geological Survey of Western Australia, 100 Plain St., East Perth, WA 6004, Australia
[c] School of Geosciences, The University of Edinburgh, The King's Buildings, West Mains Road, Edinburgh, EH9 3JW, UK
[d] Institute of the Earth's Crust, Siberian Branch of the Russian Academy of Sciences, Lermontov Str. 128, Irkutsk 664033, Russia
[e] JSL ALROSA, 12 Yuzhnaya Str., Aikhal, Republic of Sakha (Yakutia) 678190, Russia

ARTICLE INFO

Article history:
Received 26 July 2008
Received in revised form 11 January 2009
Accepted 26 January 2009

Keywords:
Geochronology
Paleomagnetism
Mafic rocks
Olenek Uplift
Siberia

ABSTRACT

We present a new, reliably dated Mesoproterozoic paleopole for Siberia, based on a combined geochronological and paleomagnetic study of mafic rocks within the Mesoproterozoic Sololi Group of the Olenek Uplift in northern Siberia. Ion microprobe (SHRIMP) U–Pb analysis yields crystallisation ages of 2036 ± 11 Ma for zircon from a basement granite and 1473 ± 24 Ma for baddeleyite from a large dolerite sill within the Kyutingde Formation. The baddeleyite result indicates that the lower Sololi Group is significantly older than was suggested by previous K–Ar results. Paleomagnetic analysis of the dolerite sill and related mafic intrusive rocks yields a paleopole at 33.6°N, 253.1°E, $A_{95} = 10.4°$. A positive baked-contact test between the Kyutingde sill and sedimentary country rocks shows that the magnetisation is primary. Comparison of this paleopole with coeval results for Laurentia provides a revised reconstruction between Siberia and Laurentia, and implies that these two continents were parts of a single Mesoproterozoic supercontinent since at least 1473 Ma. We argue that Siberia, Laurentia, and Baltica belonged to the same supercontinent between 1473 Ma and mid-Neoproterozoic time.

Crown Copyright © 2009 Published by Elsevier B.V. All rights reserved.

1. Introduction

The concept of Precambrian supercontinents has been popular in recent decades, although their constituent continental blocks, configurations, and timing are strongly debated (Rogers, 1996; Condie, 1998; Meert, 2002; Pesonen et al., 2003; Zhao et al., 2004). The wide variety of models reflects mainly the lack of reliable constraints from high-quality, well-dated paleomagnetic poles (paleopoles).

Phanerozoic Apparent Polar Wander Paths (APWPs) for the majority of continents are based on many high-quality paleopoles, and comparison of APWPs for different blocks has yielded relatively robust reconstructions and general agreement on Phanerozoic tectonic history. In contrast, Neoproterozoic and older paleomagnetic data are scarce and typically equivocal or controversial. It is impossible at this stage to construct adequate APWPs owing to time gaps of more than 100 million years between some data points, which lead to problems of longitudinal uncertainty and polarity ambiguity.

Although the recent UNESCO IGCP 440 project "Rodinia Assembly and Breakup" (Bogdanova et al., 2008) marked a significant breakthrough in our understanding of the youngest Precambrian supercontinent, Rodinia, its configuration and timing remain provisional (Dalziel, 1997; Pisarevsky et al., 2003; Li et al., 2008), owing mainly to the small number of Neoproterozoic paleomagnetic data available. The situation is worse for Mesoproterozoic and older times, for which Evans and Pisarevsky (2008) estimated the number of high-quality pre-800 Ma paleopoles at less than 50. Without sufficient paleopoles to construct adequate APWPs, it is necessary to rely primarily on the "key pole" approach (Buchan et al., 2000), which involves comparison of coeval paleopoles from different continents. Although this method can demonstrate the presence or absence of relative movements between continents, there are very few precisely coeval pairs of paleopoles available (Cawood et al., 2006).

The Siberian craton is reasonably well studied geologically, but there is a deficit of reliable Precambrian geochronology and well-dated paleopoles. This situation has improved significantly in

* Corresponding author. Present address: Geological Survey of Western Australia, 100 Plain St., East Perth, WA 6004, Australia. Tel.: +61 8 9222 3613; fax: +61 8 9222 3633.
E-mail address: michael.wingate@doir.wa.gov.au (M.T.D. Wingate).

0301-9268/$ – see front matter. Crown Copyright © 2009 Published by Elsevier B.V. All rights reserved.
doi:10.1016/j.precamres.2009.01.004

recent years (Rainbird et al., 1998; Gallet et al., 2000; Pavlov et al., 2000; Sklyarov et al., 2003; Rosen, 2002; Gladkochub et al., 2006a), leading to a better understanding of the role of Siberia in Rodinia (Pisarevsky et al., 2003, 2008). However, the pre-Rodinia history of Siberia remains vague. Of several Mesoproterozoic paleopoles in the Global Paleomagnetic Database (Pisarevsky, 2005), only the 1503 ± 5 Ma Kuonamka dykes pole (Ernst et al., 2000) is well dated.

In this contribution, we report new ion microprobe U–Pb ages for a basement granite and a large dolerite (diabase) sill in the Olenek Uplift of northern Siberia, located about 250 km south of the Laptev Sea and 100 km west of the Lena River. We also present a reliably dated Mesoproterozoic paleopole for Siberia, based on a combined geochronological and paleomagnetic study of the dolerite sill and related mafic igneous rocks. A preliminary paleomagnetic study by Gurevich (1983) suggested that mafic rocks in this area carry a stable magnetic remanence, hence they appeared to be good targets for U–Pb geochronology and paleomagnetism to obtain new constraints on Siberia's Proterozoic tectonic history.

2. Geology and sample collection

The Olenek Uplift is part of the Paleoproterozoic Olenek superterrane – one of the building blocks of the Siberian craton (Fig. 1), which was assembled between 2.1 and 1.8 Ga (Li et al., 2008; Rosen, 2002; Gladkochub et al., 2006b). The Paleoproterozoic rocks consist of folded, greenschist-grade metasediments of the Aekit Formation, intruded by post-tectonic granitoids that yielded K–Ar ages of c. 2080–2050 Ma (Krylov et al., 1963). These basement rocks are overlain unconformably by Mesoproterozoic sedimentary rocks of the Sololi Group (Fig. 1), which dip shallowly (0–12°, generally <5°) to the northwest. The lower Sololi Group consists of deltaic sediments of the Sygynakhtakh Formation, overlain by shelf clastics and carbonates of the Kyutingde Formation (Kats and Frolova, 1986). The Kyutingde Formation is intruded by mafic sills and dykes.

The largest sill, within the upper Kyutingde Formation, is composed of quartz dolerite and exhibits ophitic textures. It has a lateral extent of about 22 km, and is about 100-m thick. The Kyutingde Formation is overlain unconformably by quartz sandstone, siltstone, and carbonate sediments of the Arymas Formation, which in turn is unconformably overlain by clastic and carbonate shelf sediments of the Debengde and Haipakh formations (Fig. 1).

K–Ar dating of stromatolites within the Sololi Group by Ponomarchuk et al. (1994) suggested ages between 1440 and 900 Ma (Fig. 1). However, Gorokhov et al. (2006), based on Rb–Sr dating, proposed that mixed-layer illite-smectite was formed in shales of the Debengde Formation during two periods, at c. 1272–1211 and 1080–1038 Ma. They suggested that the first interval reflects burial diagenesis and approximates the time of sediment

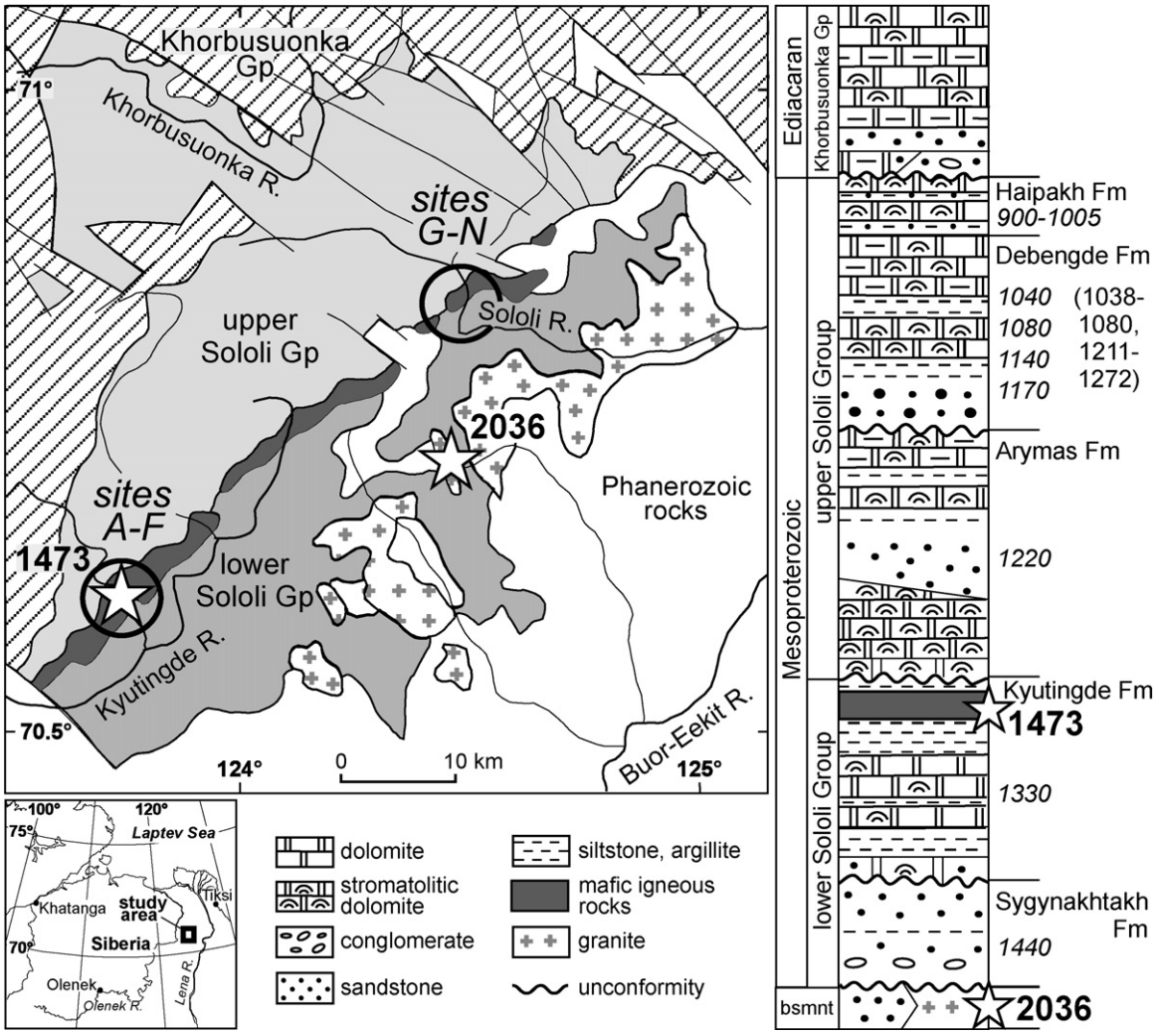


Fig. 1. Simplified geology and stratigraphy of the eastern Olenek Uplift and sampling areas of this study. Numbers in italics are K–Ar dates (in Ma) of Ponomarchuk et al. (1994). Numbers in brackets for Debengde Formation are Rb–Sr illite ages (in Ma) of Gorokhov et al. (2006). Stars indicate locations of rocks dated in this study (numbers in bold font are U–Pb dates in Ma).

deposition, whereas the second may be related to regional uplift and changes in the hydrological regime during the pre-Haipakh Formation break in sedimentation. If a 1272–1211 Ma age for the Debengde Formation is accepted, the underlying units (Arymas, Kyutingde, and Sygynakhtakh Formations) may be significantly older than is implied by the stromatolite K–Ar results.

However, Khomentovsky (2006) argued, with reference to Faure (1986, p. 130), that Rb–Sr dating of illite grains of >2 μm size may reflect the age of the source regions from which the coarse illite originated. Because the 1272–1211 Ma illites of Gorokhov et al. (2006) ranged from 0.6 to 5 μm in size, they may represent a mixture of authigenic and detrital grains, whereas the 1038–1080 Ma illites were <0.1 μm and may instead reflect diagenesis of the Debengde Formation. In addition, Stanevich et al. (2009) recently found microfossils in the Debengde Formation that may belong to *aff. Glomovertella Reitl.* and *aff. Obruchevella Reitl*, which so far have been found only in Neoproterozoic successions. Therefore, taking a conservative approach, we consider the Arymas and Debengde formations to have been deposited prior to about 1000 Ma.

Samples for geochronology were collected from two sites in the large dolerite sill within the Kyutingde Formation, and one from a basement granite (Fig. 1). Paleomagnetic samples were drilled from 14 sites in 2 localities (Fig. 1). The Kyutingde dolerite sill and several dolerite dykes, as well as an exposure of sedimentary rocks underlying the sill, were sampled near the Kyutingde River (sites A–F), and several fine-grained sills and dykes were sampled along and near the Sololi River (sites G–N). The rocks in both areas are essentially flat-lying; the Kyutingde sill is concordant with bedding in underlying sedimentary rocks, and the attitude of the Sololi rocks is indicated locally by amygdaloidal horizons. A stable magnetisation was reported by Gurevich (1983) for a single site in mafic rocks along the Sololi River. Geochemical data confirm a genetic relationship between all the studied rocks, although those results will be presented in a separate paper.

3. Geochronology

Zircon and baddeleyite crystals were separated from crushed samples using a Wilfley table and standard magnetic and high-density liquid methods, followed by hand-picking under a binocular microscope. Crystals were cast, together with chips of a zircon reference standard, in an epoxy disc, which was polished to expose the interiors of the crystals. All crystals were imaged using transmitted and reflected light microscopy and cathodoluminescence (CL) techniques. The sample mount was then cleaned thoroughly, then vacuum-coated with ~50 nm Au and loaded into the SHRIMP sample lock to pump down to high vacuum for 24 h prior to analysis.

Measurements of U, Th, and Pb were conducted using the SHRIMP II ion microprobe at Curtin University of Technology in Perth, Australia. U–Th–Pb ratios and absolute abundances were determined relative to the CZ3 zircon standard ($^{206}Pb/^{238}U = 0.09143$ (564 Ma), 551 ppm ^{238}U; Pidgeon et al., 1984; Nelson, 1997), analyses of which were interspersed with those of unknown grains, using operating procedures similar to those described by Claoue-Long et al. (1995). Data were reduced using SQUID (Ludwig, 2001), and decay constants recommended by Steiger and Jäger (1977). Measured compositions in zircon were corrected for common Pb using non-radiogenic ^{204}Pb. Baddeleyite analyses were corrected for common Pb by the 208-method (Compston et al., 1984; Wingate et al., 1998). In both cases, an average crustal composition (Stacey and Kramers, 1975) appropriate to the age of the mineral was assumed. Ratios of $^{238}U/^{206}Pb$ measured

Table 1
U–Pb analytical data for zircon from basement granite sample 03138.

Grain area	^{238}U	^{232}Th	Th/U	f	$^{238}U/^{206}Pb$		$^{207}Pb/^{206}Pb$		$^{238}U/^{206}Pb$ age		$^{207}Pb/^{206}Pb$ age		Disc.
	(ppm)	(ppm)		(%)		(±1σ)		(±1σ)	(Ma)	(±1σ)	(Ma)	(±1σ)	(%)
1.1	1068	153	0.15	0.315	3.916	0.039	0.11356	0.00061	1466	13	1857	10	21.1
1.2	699	165	0.24	0.081	2.676	0.073	0.12505	0.00083	2047	48	2030	12	−0.9
2.1	304	73	0.25	0.627	3.200	0.032	0.12756	0.00130	1753	15	2065	18	15.1
3.1	2546	136	0.06	0.209	4.829	0.044	0.10101	0.00042	1213	10	1643	8	26.1
3.2	212	50	0.24	0.035	3.196	0.035	0.12375	0.00076	1755	17	2011	11	12.7
4.1	4046	425	0.11	0.909	22.71	0.235	0.08945	0.00110	278	3	1414	23	80.3
5.1	2581	107	0.04	0.629	8.708	0.092	0.09123	0.00061	701	7	1451	13	51.7
6.1	463	279	0.62	0.636	4.869	0.095	0.12726	0.00096	1204	21	2060	13	41.6
7.1	487	156	0.33	1.084	3.500	0.035	0.12626	0.00413	1620	14	2047	58	20.8
7.2	175	40	0.24	0.498	2.996	0.036	0.13080	0.00160	1857	19	2109	21	12.0
8.1	464	98	0.22	0.908	4.761	0.047	0.12671	0.00114	1229	11	2053	16	40.1
9.1	190	115	0.63	0.201	2.886	0.030	0.12759	0.00091	1918	17	2065	13	7.1
9.2	2680	215	0.08	1.216	11.42	0.118	0.08978	0.00141	541	5	1421	30	61.9
9.3	226	116	0.53	0.564	3.660	0.043	0.12849	0.00150	1557	16	2077	21	25.0
10.1	1424	384	0.28	0.416	5.049	0.047	0.09906	0.00061	1165	10	1607	11	27.5
11.1	911	171	0.19	0.811	8.383	0.147	0.11874	0.00111	727	12	1937	17	62.5
12.1	120	52	0.45	−0.023	2.757	0.031	0.12576	0.00082	1995	19	2040	12	2.2
12.2	133	54	0.42	0.093	2.754	0.030	0.12513	0.00084	1997	19	2031	12	1.7
12.3	125	54	0.45	−0.026	2.731	0.033	0.12606	0.00081	2012	21	2044	11	1.6
13.1	371	40	0.11	1.227	3.321	0.051	0.12716	0.00690	1697	23	2059	96	17.6
14.1	249	42	0.17	0.437	3.079	0.031	0.12549	0.00092	1813	16	2036	13	10.9
15.1	262	112	0.44	0.607	3.127	0.032	0.12574	0.00120	1789	16	2039	17	12.3
15.2	396	141	0.37	1.002	4.029	0.040	0.12487	0.00124	1429	13	2027	18	29.5
16.1	142	139	1.01	0.433	3.936	0.043	0.12410	0.00131	1460	14	2016	19	27.6
16.2	292	220	0.78	0.626	4.307	0.069	0.11927	0.00260	1346	20	1945	39	30.8
17.1	67	41	0.63	−0.044	2.699	0.035	0.13114	0.00116	2032	23	2113	16	3.9
17.2	69	42	0.63	−0.002	2.744	0.035	0.13016	0.00118	2003	22	2100	16	4.6
17.3	66	36	0.55	0.442	2.958	0.039	0.12825	0.00172	1877	22	2074	24	9.5
18.1	2471	373	0.16	1.474	6.616	0.069	0.10357	0.00153	907	9	1689	27	46.3
19.1	235	74	0.33	0.707	2.654	0.027	0.13254	0.00193	2062	18	2132	25	3.3
20.1	816	146	0.19	1.724	5.833	0.042	0.12200	0.00238	1020	7	1986	35	48.6
21.1	1398	581	0.43	0.294	6.134	0.033	0.09098	0.00075	974	5	1446	16	32.7
22.1	1360	194	0.15	0.222	4.624	0.831	0.09546	0.00062	1262	206	1537	12	17.9

Notes: f is the proportion of common ^{206}Pb in measured ^{206}Pb. Isotope ratios and ages are corrected for common Pb using measured $^{204}Pb/^{206}Pb$.

in baddeleyite by ion microprobe are biased by crystal orientation effects (Wingate and Compston, 2000), hence reliable ages are based on $^{207}Pb/^{206}Pb$ ratios. However, these ratios in baddeleyite were calibrated against the CZ3 zircon standard as a means of qualitatively assessing loss of radiogenic Pb. Weighted mean ages are quoted below with 95% confidence intervals.

3.1. Basement granite (sample 03138)

Sample 03138 is a coarse-grained biotite granite, which is relatively fresh apart from minor chloritisation of biotite and seritisation of plagioclase. The sample yielded abundant subhedral to euhedral zircons up to 400-μm long. Most are equant in shape, although a few elongate crystals have aspect ratios up to 8:1. They range from clear and colourless to brown and turbid. Concentric growth zoning is common; no obvious older cores could be discerned in CL images. Thirty-three analyses of 22 zircons were conducted (Table 1). Uranium concentrations vary widely, from 66 to 4046 ppm, and Th varies from 36 to 581 ppm, with Th/U ratios between 0.04 and 1.01. The proportion of common ^{206}Pb in total measured ^{206}Pb (f in Table 1) averages 0.5%.

The analyses range from concordant to strongly discordant (Fig. 2a). Apparent ages decrease with increasing U content, indicating that discordance is due to Pb loss related to radiation damage (metamictisation). Only seven analyses of four crystals are less than 5% discordant, and these define two age components (Fig. 2b): a concordant grouping at 2036 ± 11 Ma (n = 4, MSWD = 0.35), and a second (4% discordant) at 2111 ± 20 Ma (n = 3, MSWD = 0.58). We interpret the younger concordant result of 2036 ± 11 Ma to represent the age of granite crystallisation, and consider the older crystals to be inherited.

3.2. Kyutingde sill (samples GAB1 and 03165)

Sample GAB1 consists of coarse-grained quartz dolerite, dominated by clinopyroxene and plagioclase with ophitic texture. Accessory minerals include apatite, ilmenite, titanomagnetite, zircon, and baddeleyite. Pyroxene exhibits minor chloritisation and plagioclase shows minor sericite alteration. Four kilograms of rock yielded 75 zircons and only four baddeleyite crystals (it is likely that many baddeleyite grains were lost during sample processing). It was hoped that the zircons were of a primary origin and could be used to date the crystallisation of the sill. The zircons range from clear and colourless to dark brown and opaque. Most are subhedral to euhedral, slightly rounded, and range up to 250 μm in length with aspect ratios up to 5:1. Concentric growth zoning is present in all zircons. A total of 31 analyses were conducted of 24 zircons (Table 2). Concentrations of ^{238}U range from 140 to 2193 ppm, ^{232}Th from 38 to 1085 ppm, and Th/U from 0.12 to 2.1. Values for f vary between 0.04 and 4.8%, with a median of 0.9%.

The zircons have undergone large amounts of radiogenic Pb loss, and most analyses are strongly discordant (Fig. 2c). Both $^{207}Pb/^{206}Pb$ and $^{238}U/^{206}Pb$ ages decrease with increasing U content, indicating that Pb loss reflects radiation damage. Considering only near-concordant data (<7% discordant), there are 3 groups of $^{207}Pb/^{206}Pb$ ages (all ±1σ): 2564 ± 11 Ma (2 analyses of 1 zircon,

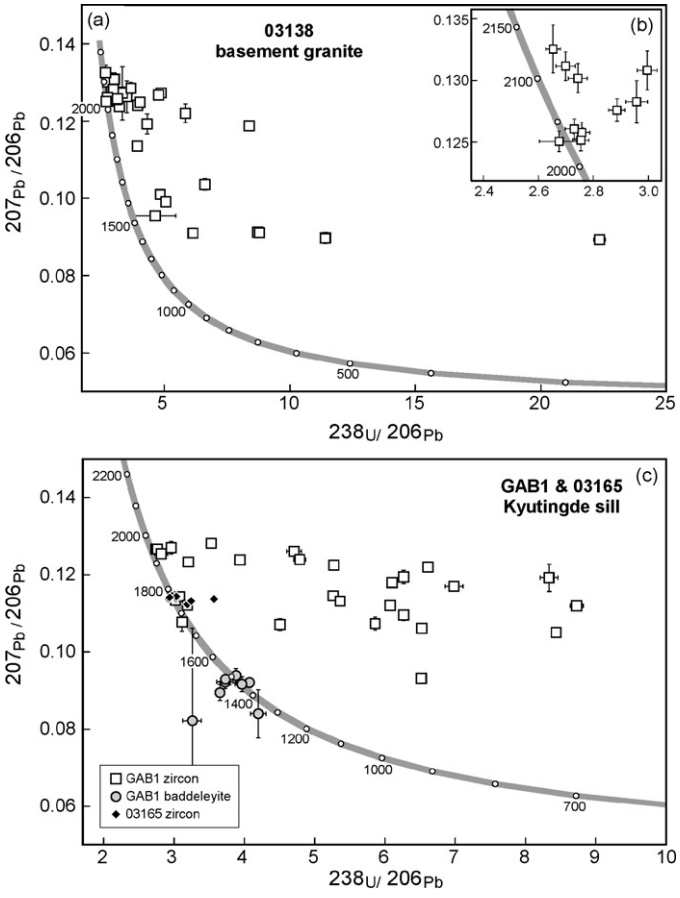


Fig. 2. U–Pb analytical data for dated samples: (a) basement granite sample 03138, (b) expanded view of near-concordant data for sample 03138, (c) dolerite sill samples GAB1 and 03165 (two concordant analyses of a single zircon at 2564 Ma are not shown). All data are corrected for common Pb; error bars are 1σ.

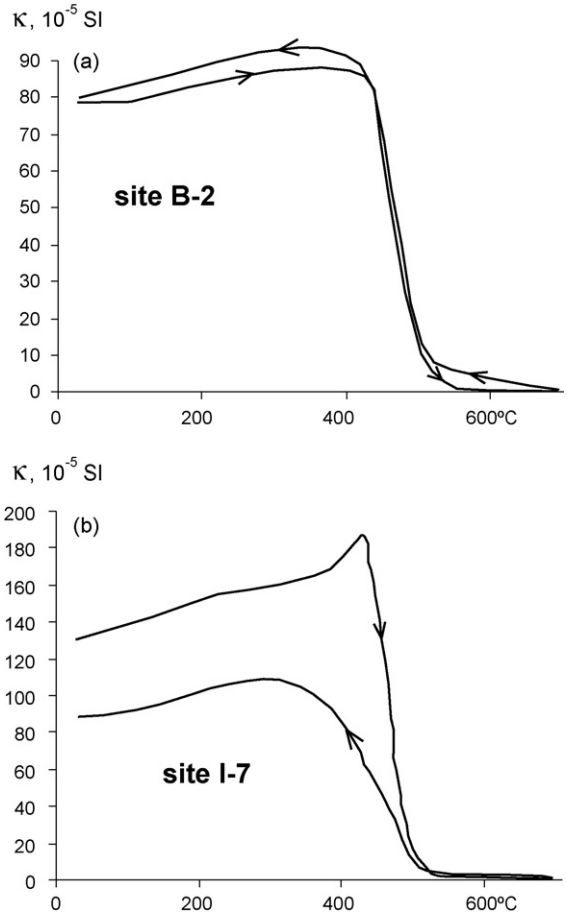


Fig. 3. Analysis of Curie points using magnetic susceptibility versus temperature curves: (a) Kyutingde area; (b) Sololi area.

Table 2
U–Pb analytical data for zircon and baddeleyite from sill samples GAB1 and 03165.

Grain area	^{238}U	^{232}Th	Th/U	f	^{238}U/^{206}Pb		^{207}Pb/^{206}Pb		^{238}U/^{206}Pb age		^{207}Pb/^{206}Pb age		Disc.
	(ppm)	(ppm)		(%)		(±1σ)		(±1σ)	(Ma)	(±1σ)	(Ma)	(±1σ)	(%)
GAB1 zircon													
1.1	382	223	0.60	0.042	2.068	0.028	0.17091	0.00072	2543	29	2567	7	0.9
1.2	384	253	0.68	0.083	2.080	0.026	0.17020	0.00094	2531	27	2560	9	1.1
2.1	234	475	2.10	2.286	3.021	0.036	0.11342	0.00267	1844	19	1855	43	0.6
2.2	229	470	2.12	2.358	3.114	0.035	0.10773	0.00399	1795	18	1761	68	−1.9
3.1	376	217	0.60	1.849	4.788	0.087	0.12398	0.00172	1223	20	2014	25	39.3
4.1	141	77	0.56	0.113	2.739	0.030	0.12668	0.00090	2006	19	2052	13	2.2
4.2	225	134	0.61	0.537	2.817	0.097	0.12541	0.00170	1959	58	2035	24	3.7
5.1	679	201	0.31	0.765	6.982	0.131	0.11701	0.00146	863	15	1911	22	54.8
6.1	168	46	0.28	1.394	4.707	0.102	0.12606	0.00228	1242	25	2044	32	39.2
6.2	141	38	0.28	0.153	2.963	0.044	0.12701	0.00185	1875	24	2057	26	8.9
7.1	249	93	0.39	0.178	3.530	0.041	0.12815	0.00134	1608	16	2073	18	22.4
8.1	455	915	2.08	0.347	3.079	0.032	0.11435	0.00067	1813	17	1870	11	3.0
8.2	372	519	1.44	0.255	3.190	0.026	0.11210	0.00100	1758	13	1834	16	4.1
9.1	1665	490	0.30	0.769	6.075	0.063	0.11203	0.00087	982	9	1833	14	46.4
10.1	937	759	0.84	2.009	8.334	0.132	0.11922	0.00397	730	11	1945	60	62.4
10.2	605	321	0.55	0.882	6.102	0.050	0.11796	0.00179	978	7	1926	27	49.2
11.1	837	380	0.47	0.885	6.524	0.046	0.10613	0.00150	919	6	1734	26	47.0
12.1	530	151	0.29	4.826	6.265	0.061	0.11941	0.00509	955	9	1947	76	51.0
13.1	2193	729	0.34	0.553	8.436	0.071	0.10502	0.00176	722	6	1715	31	57.9
14.1	1091	195	0.18	2.243	5.860	0.039	0.10737	0.00279	1016	6	1755	48	42.1
15.1	1938	1086	0.58	2.009	8.727	0.094	0.11192	0.00215	699	7	1831	35	61.8
16.1	720	391	0.56	1.951	6.610	0.051	0.12201	0.00266	908	7	1986	39	54.3
17.1	307	117	0.39	0.533	3.202	0.030	0.12331	0.00147	1752	14	2005	21	12.6
18.1	949	329	0.36	0.689	2.757	0.022	0.12669	0.00095	1995	14	2052	13	2.8
19.1	991	303	0.32	2.335	6.266	0.049	0.10955	0.00373	955	7	1792	62	46.7
20.1	726	163	0.23	1.561	5.362	0.038	0.11314	0.00210	1102	7	1851	34	40.4
21.1	505	797	1.63	1.898	5.273	0.051	0.12248	0.00314	1120	10	1993	46	43.8
22.1	363	229	0.65	0.728	3.941	0.036	0.12387	0.00175	1458	12	2013	25	27.6
23.1	744	417	0.58	2.234	5.261	0.071	0.11454	0.00222	1122	14	1873	35	40.1
24.1	2156	262	0.13	0.833	6.518	0.042	0.09313	0.00098	920	6	1491	20	38.3
25.1	907	328	0.37	3.987	4.509	0.030	0.10708	0.00346	1291	8	1750	59	26.2
GAB1 baddeleyite													
1.1	117	8	0.07	0.282	3.725	0.117	0.09225	0.00171	1533	43	1472	35	−3.1
1.2	26	2	0.08	2.687	4.198	0.109	0.08398	0.00626	1377	32	1292	145	−6.6
1.3	89	4	0.05	0.134	3.733	0.047	0.09282	0.00098	1530	17	1484	20	−3.0
1.4	83	4	0.05	0.263	3.883	0.067	0.09382	0.00185	1477	23	1505	37	1.8
1.5	62	3	0.05	0.058	3.653	0.066	0.08953	0.00205	1560	25	1415	44	−10.2
2.1	87	9	0.10	0.489	4.073	0.048	0.09200	0.00114	1415	15	1467	24	3.5
2.2	80	6	0.08	0.927	3.967	0.098	0.09164	0.00193	1449	32	1460	40	0.7
3.1	44	3	0.08	8.175	3.258	0.133	0.08218	0.02397	1725	62	1250	571	−38.0
03165 zircon													
1.1	126	81	0.66	0.183	3.569	0.026	0.11357	0.00093	1592	10	1857	15	14.3
1.2	235	70	0.31	0.117	3.243	0.020	0.11309	0.00052	1733	9	1850	8	6.3
2.1	267	92	0.35	0.065	3.032	0.026	0.11373	0.00045	1838	14	1860	7	1.2
3.1	175	61	0.36	0.124	2.927	0.020	0.11365	0.00060	1895	11	1859	9	−1.9
3.2	300	85	0.29	0.026	3.033	0.020	0.11427	0.00039	1837	11	1868	6	1.7
4.1	236	96	0.42	−0.011	2.976	0.017	0.11446	0.00043	1867	9	1871	7	0.2
4.2	188	79	0.44	0.000	2.941	0.026	0.11490	0.00055	1887	14	1878	9	−0.5
5.1	252	150	0.62	0.129	3.189	0.018	0.11221	0.00054	1758	9	1836	9	4.2
6.1	147	51	0.36	0.025	2.939	0.020	0.11405	0.00071	1888	11	1865	11	−1.2

Notes: Isotope ratios and ages for baddeleyite are corrected for common Pb using measured ^{208}Pb/^{206}Pb and Th/U (208-method). Other notes as in Table 1.

MSWD = 0.36), 2053 ± 17 Ma (3 analyses of 3 zircons, MSWD = 0.04), and 1857 ± 37 Ma (4 analyses of 2 zircons, MSWD = 1.84).

Of the four baddeleyite crystals, only three were of sufficient size to analyse, and eight analyses were obtained (Table 2). Concentrations of ^{238}U are relatively low, and range from 26 to 117 ppm, and ^{232}Th varies from 2 to 8 ppm, with Th/U ratios between 0.05 and 0.1. Values for f vary from 0.05 to 8.2%, with a median of 0.39%. Six analyses with the lowest common Pb content (f < 1%) yield ^{207}Pb/^{206}Pb ratios that agree to within analytical uncertainty and yield a weighted mean age of 1473 ± 24 Ma (MSWD = 0.58). The two excluded analyses (1.2, 3.1), with the highest common Pb (f = 2.7 and 8.2%), indicated imprecise ^{207}Pb/^{206}Pb ages of 1292 and 1250 Ma. Calibration of baddeleyite ^{238}U/^{206}Pb ratios against the CZ3 zircon standard shows qualitatively that the six accepted analyses are approximately concordant (Fig. 2). Together with the relatively low dispersion of ^{207}Pb/^{206}Pb ratios, this indicates that loss of radiogenic Pb is negligible.

Dolerite sample 03165 yielded eight subhedral, rounded zircons, and nine analyses were obtained from six crystals (Table 2). Uranium concentrations vary from 126 to 300 ppm, ^{232}Th varies from 51 to 150 ppm, and Th/U ratios are well grouped between 0.29 and 0.66. Values for f are less than 0.2% and average 0.07%. The analyses range from concordant to moderately discordant. Six analyses less than 2% discordant yield a weighted mean ^{207}Pb/^{206}Pb age of 1867 ± 6 Ma (MSWD = 0.81). Three excluded analyses are 4–14% discordant.

Baddeleyite inheritance in mafic intrusions is very unlikely (Krogh et al., 1987; Heaman and Lecheminant, 1993), hence we interpret the mean ^{207}Pb/^{206}Pb age of 1473 ± 24 Ma for baddeleyite from sample GAB1 as the age of crystallisation of the Kyutingde

dolerite sill. Consequently, we consider all the zircons analysed from samples GAB1 and 03165, and the older zircons in sample 03138, to be xenocrysts. The xenocrysts fall into four age groups: 2564, 2111, 2053, and ~1865 Ma. The 2053 ± 17 Ma result for GAB1 is similar to the age of 2036 ± 11 Ma obtained for basement granite sample 03138. The 2564, 2111, and c. 1865 Ma results are previously unrecognised in the Olenek Uplift and presumably indicate that rocks of this age are present in the region.

4. Paleomagnetism

Remanence composition was determined by detailed stepwise thermal demagnetisation (≤20 steps, to 630 °C), using a *Magnetic Measurements* thermal demagnetiser and a *2G-755R* cryogenic magnetometer. Stepwise alternating field (AF) demagnetisation (≤26 steps, up to 160 mT) was also applied using a *2G-600* automated degaussing system. To monitor possible mineralogical changes during heating, magnetic susceptibility was measured in selected samples after each heating step using a *Bartington MS2* susceptibility meter. Magnetic mineralogy was investigated from demagnetisation characteristics and, in selected samples, from detailed variation of susceptibility versus temperature (20–700 °C) obtained using the *Bartington* meter in conjunction with an automated *Bartington* furnace.

Natural remanent magnetisation (NRM) intensities of the mafic rocks range from 0.1 to 8.5 A/m, and their magnetic susceptibilities

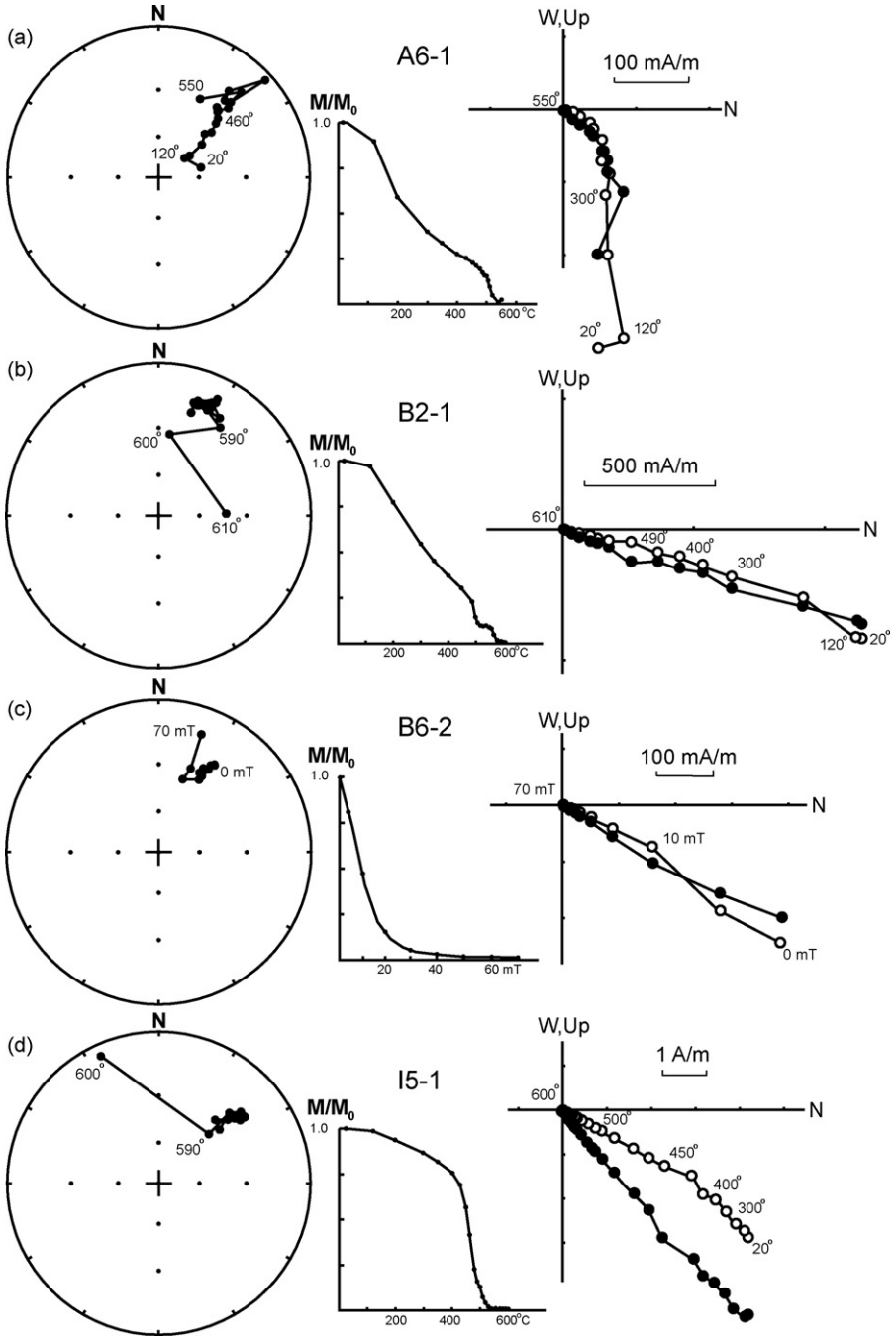


Fig. 4. Stereoplots and representative examples of demagnetisation behaviour. In orthogonal plots, open (closed) symbols show magnetisation vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetisation. Stereoplots show upwards (downwards) pointing paleomagnetic directions with open (closed) symbols. (a–c) Kyutingde area, site A and B; (d) Sololi area, site I.

Table 3
Site-mean paleomagnetic directions for mafic rocks of the Kyutingde–Sololi area.

Site	*N*/*n*	Site location		*D* (°)	*I* (°)	*k*	α_{95} (°)	Paleopole		*Dp* (°)	*Dm* (°)
		Latitude (°N)	Longitude (°E)					Latitude (°N)	Longitude (°E)		
A	7/12	70.5909	123.7525	46.8	32.1	35.0	10.4	30.0	250.4	6.6	11.7
B	6/10	70.5899	123.7415	31.7	26.4	41.5	10.5	30.1	267.6	6.2	11.4
C	4/8	70.6004	123.7120	55.8	16.4	124.8	8.3	18.8	243.9	4.4	8.6
D	6/12	70.6007	123.7174	54.3	37.2	16.9	16.8	31.1	241.3	11.6	19.7
F	4/7	70.5499	123.8449	21.8	54.6	71.4	10.9	52.7	273.8	10.9	15.4
G	8/17	70.8527	124.5320	53.4	24.9	34.8	9.5	23.8	245.8	5.5	10.2
H	10/19	70.8518	124.5298	58.4	22.2	24.7	9.9	20.9	241.2	5.5	10.5
I	7/19	70.8475	124.5301	53.3	18.7	314.9	3.4	20.5	246.9	1.8	3.5
J	7/17	70.8463	124.5313	54.0	17.2	139.7	5.1	19.6	246.5	2.7	5.3
K	4/11	70.8396	124.5362	19.5	50.7	38.1	15.1	49.2	278.7	13.7	20.3
L	3/4	70.8211	124.4697	36.4	21.9	13.3	35.3	26.4	263.9	19.7	37.3
M	6/11	70.8211	124.4697	39.1	33.6	21.9	14.6	32.7	259.2	9.5	16.6
N	6/12	70.8219	124.4770	28.9	76.3	21.1	14.9	77.2	232.0	25.5	27.6
Mean of 13 sites (A–N, excluding site E)								33.6	253.1	A_{95} = 10.4°	
E	9/12	70.5659	123.7956	316.2	2.1	6.8	21.2	14.9	349.5	10.6	21.2

Notes: *N*/*n*, number of samples/specimens; *D*, *I*, site-mean declination, inclination; *k*, precision parameter of Fisher (1953); α_{95}, A_{95}, semi-angle of 95% cone of confidence about mean direction, pole; *Dp*, *Dm*, semi-axes of the cone of 95% confidence about the pole. Site E is in sedimentary rocks of the Kyutingde Formation.

from about 0.01 to 0.08 SI units. Susceptibility versus temperature curves (Fig. 3) show that Curie points are distributed between 450 and 570 °C, indicative of low-titanium titanomagnetite as the main magnetic carrier. Several curves exhibit a Hopkinson peak (e.g. Fig. 3b), indicating that the remanence is carried in part by highly stable single-domain particles.

In many cases, both thermal and AF demagnetisation isolated a single, stable, unipolar remanence component directed shallowly to moderately downward to the NE (Fig. 4). There are also indications (e.g. Fig. 4a and b) of a low-stability, steeply downward component dispersed around the present-day dipole field (PDF) direction. Several samples from site A (high-standing cliff outcrops) possess a strong (>4 A/m) low-coercivity, randomly oriented remanence, probably caused by lightning strikes. These latter data were excluded from further consideration.

Site-mean directions for the mafic rocks are listed in Table 3 and illustrated in Fig. 5a. Because the majority of rocks in the area are essentially flat-lying, no tectonic corrections have been applied to the results. The overall mean pole of 33.6°N, 253.1°E (A_{95} = 10.4°) is compatible with the preliminary VGP (Table 4) obtained previously (Gurevich, 1983) from a single site on the Sololi River.

4.1. Baked-contact test

At site B, the lower contact of the Kyutingde sill is exposed. Underlying sedimentary rocks have been thermally metamorphosed by the sill, and range from crystalline marble at the contact to weakly hornfelsed limestone more than 20 m below the contact. These rocks were sampled for a paleomagnetic baked-contact test to determine if the sill remanence is primary. Samples were also collected, at site E, from sedimentary rocks which are stratigraphically further below, and mostly unaffected by, the dolerite sill.

Following removal of a low-stability viscous component, thermal demagnetisation of baked sediments within 20 m of the contact isolated a single stable remanence component with a direction close to that in site B dolerite (Fig. 6a and f). Two samples collected more than 20 m below the sill show more complex demagnetisation behaviour (Fig. 6b) with a mid-temperature PDF component and scattered high-temperature shallow NW component (Fig. 6b and f). Only one specimen (Fig. 6c) collected at 13.5 m from the sill show evidence of the NW component together with the "sill" component. Unfortunately, this specimen disintegrated during heating to 600 °C and could not be reconstructed.

Thermal demagnetisation of mainly non-metamorphosed sedimentary rocks of the Kyutingde Formation at site E reveals a bipolar, scattered NW–SE shallow remanence (Fig. 6d and e; Table 3). Some specimens from site E were also partly demagnetised by the sill, but small remnants of the NW–SE magnetisation survived (Fig. 6e). The mean direction of this component is significantly different from

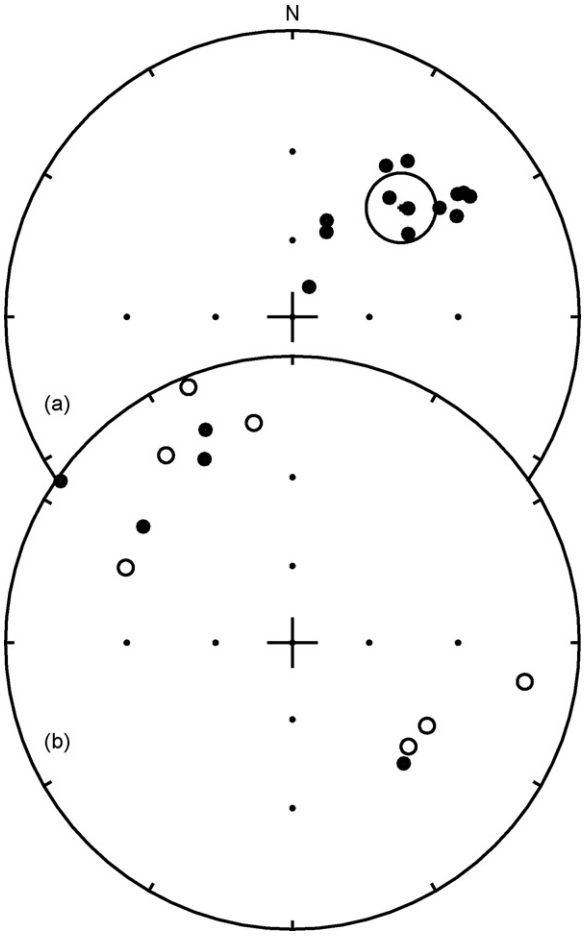


Fig. 5. (a) Site-mean directions for the Kyutingde–Sololi sill; (b) sample-mean directions for unbaked sediments of the Kyutingde Formation at site E.

Table 4
Selected Mesoproterozoic paleopoles for Siberia.

Pole	Rock unit	Age (Ma)	Latitude (°N)	Longitude (°E)	*Dp* (°)	*Dm* (°)	Reference	Result[a]
Fo	Fomich River dykes	1513 ± 51	19.2	257.8	3.0	5.9	1	–
Ku	Kuonamka dykes	1503 ± 5	6.0	234.0	14.0	28.0	2	8554
Kt	Kyutingde–Sololi mafic rocks	1473 ± 24	33.6	253.1	10.4	10.4	This study	–
So	Sololi mafic rocks (VGP)	1473 ± 24[b]	23.0	231.0	2.5	4.4	3	5657
Ar	Arymas Formation	<1473?	12.0	250.0	5.0	10.0	4	5656
Ch	Chieress dyke (VGP)	1384 ± 2	4.0	258.0	5.0	9.0	2	8555

Notes: Pole, abbreviated pole names are those used in Fig. 7. VGP, virtual paleomagnetic pole. See also notes of Table 3. References: 1. Veselovsky et al. (2006), 2. Ernst et al. (2000), 3. Gurevich (1983), 4. Iosifidi and Rodionov (1986).

[a] Result number is that used in the Global Paleomagnetic Database v. 4.6 (Pisarevsky, 2005).

[b] Age is taken from this study.

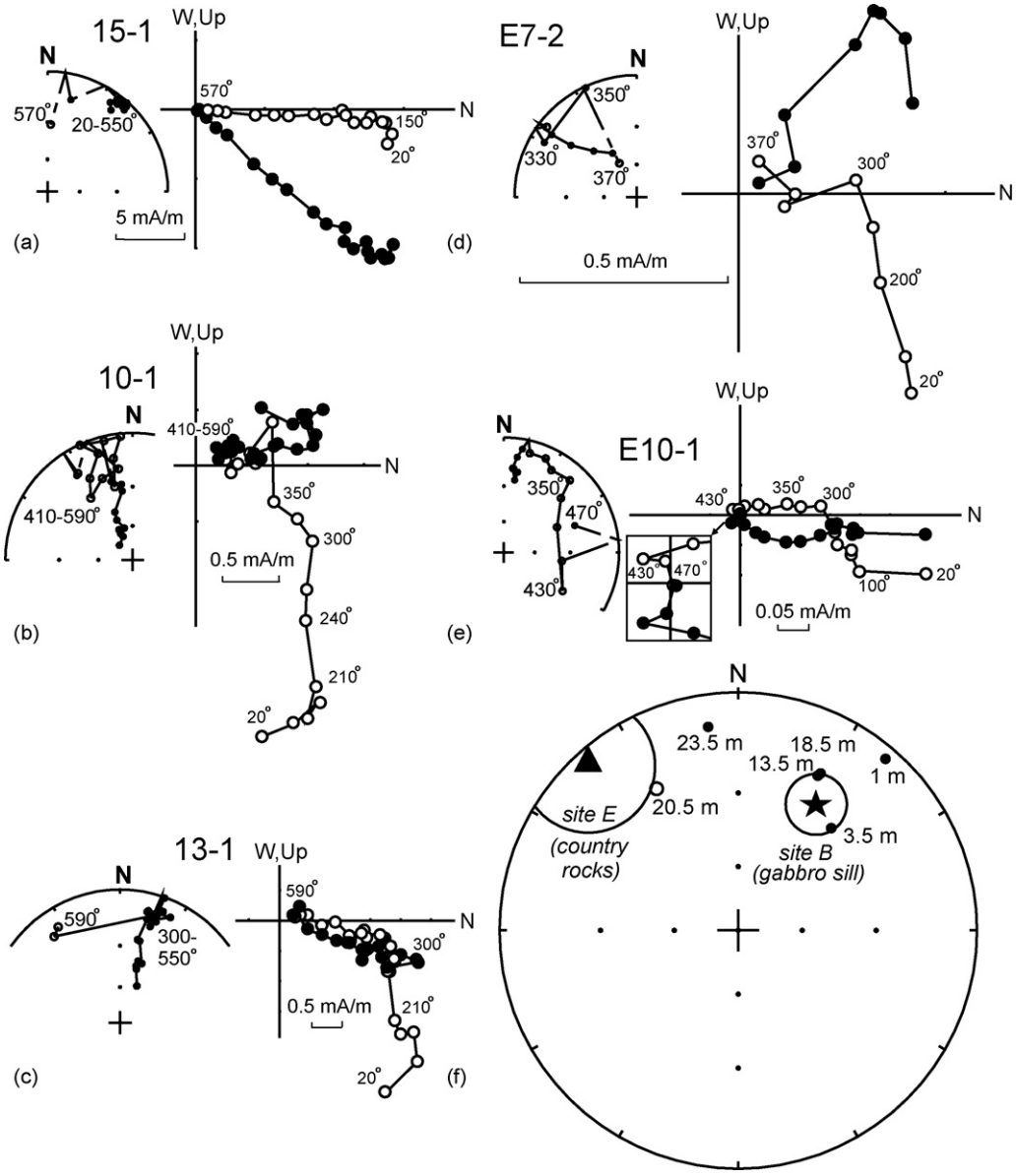


Fig. 6. Baked-contact test: demagnetisation behaviour of samples (a) <1 m, (b) >20 m, (c) 13.5 m from the lower sill contact at site B; (d, e) unbaked sedimentary country rocks at site E; (f) stereoplot for the country rocks with distances from the sill in meters at site B. Star denotes the mean direction for site B; triangle denotes the mean direction for site E; both are shown with α95 confidence circles. Other notes as in Fig. 4.

the "sill" direction. It is clear that the baked-contact test is positive (Fig. 6f) and that the remanence of the Kyutingde sill, and related mafic rocks, is primary and dates from the time of mafic magmatism at 1473 ± 24 Ma.

5. Discussion

5.1. Ages of the Kyutingde sill and Sololi group

The U–Pb baddeleyite age of 1473 Ma for the Kyutingde sill shows that the Kyutingde Formation is of at least this age, and that the previous K–Ar stromatolite dates (Ponomarchuk et al., 1994) of 1330 and 1440 Ma (Fig. 1) for the Kyutingde and Sygynakhtakh Formations are too young. Although the precise ages of the Kyutingde and Sygynakhtakh Formations remain unclear, the difference in paleomagnetic directions (Fig. 5 and Table 3) indicates that these sedimentary units are significantly older than the mafic rocks which intrude them.

Iosifidi and Rodionov (1986) reported a paleopole for the Arymas Formation (Ar, Table 4 and Fig. 7) which is reasonably close to our new result, suggesting that the deposition of the Arymas Formation may have occurred shortly after ~1470 Ma, or possibly that the sampled rocks in the Arymas Formation were remagnetised by the sill. On the other hand, our pole for the Kyutingde Formation (Site E, Table 3), although imprecise, is significantly different from poles for both the Kyutingde–Sololi mafic rocks and Arymas Formation (Kt and Ar), indicating that the time gap between the Kyutingde and Arymas Formations (Fig. 1) may have been quite long. We do not employ our result for the Kyutingde Formation (site E) for tectonic applications because of its low precision (Table 3) and negative reversal test (McFadden and McElhinny, 1990).

5.2. Mesoproterozoic reconstructions of Siberia and Laurentia

The 1473 Ma paleopole for the Kyutingde–Sololi mafic rocks (Kt, Table 4 and Fig. 7) helps to fill a significant gap in the paleomagnetic record for Siberia. The interval between ~950 and 1050 Ma is constrained paleomagnetically by a series of paleopoles (Fig. 7) from the Uchur-Maya area (Gallet et al., 2000; Pavlov et al., 2000, 2002). The Laurentian APWP has a similar shape for this interval, hence we

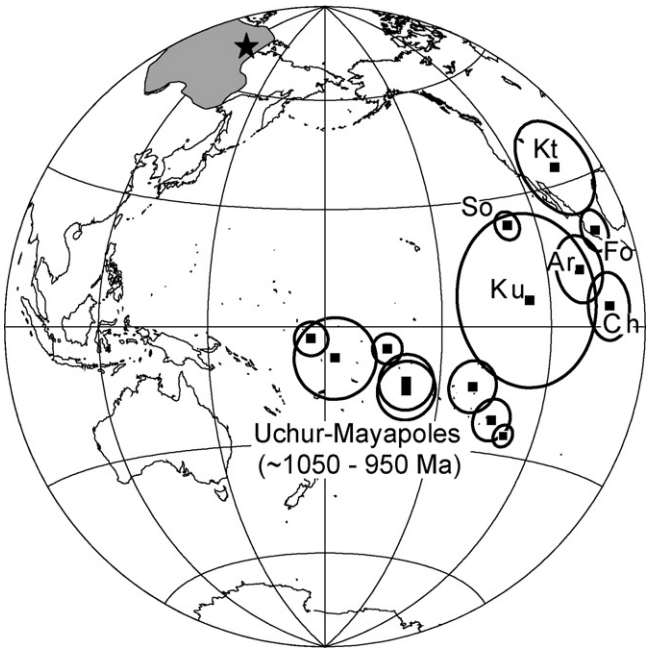


Fig. 7. Mesoproterozoic paleopoles for Siberia. See Table 4 in this paper and Table 2 in Pisarevsky and Natapov (2003). Siberia is shaded; study location is indicated by a star.

suggest that two continents were parts of the same supercontinent in the latest Mesoproterozoic–early Neoproterozoic. The best fit between the Siberian and Laurentian APWPs indicates that a significant separation existed between Laurentia and Siberia (Pisarevsky and Natapov, 2003). This space may have been occupied by another continent, or continents (Arctida of Zonenshain et al. (1990) is one possible candidate). However, it is not clear for how long this continent configuration existed prior to 1050 Ma.

The 1476 ± 16 Ma (U–Pb) St. Francois pole of Meert and Stuckey (2002) of 13.2°S, 219°E (dp/dm = 4.7°/8.0°) for Laurentia is the same age (within uncertainty) as our result for the Kyutingde–Sololi mafic rocks. Comparing this pair of coeval poles permits us to deter-

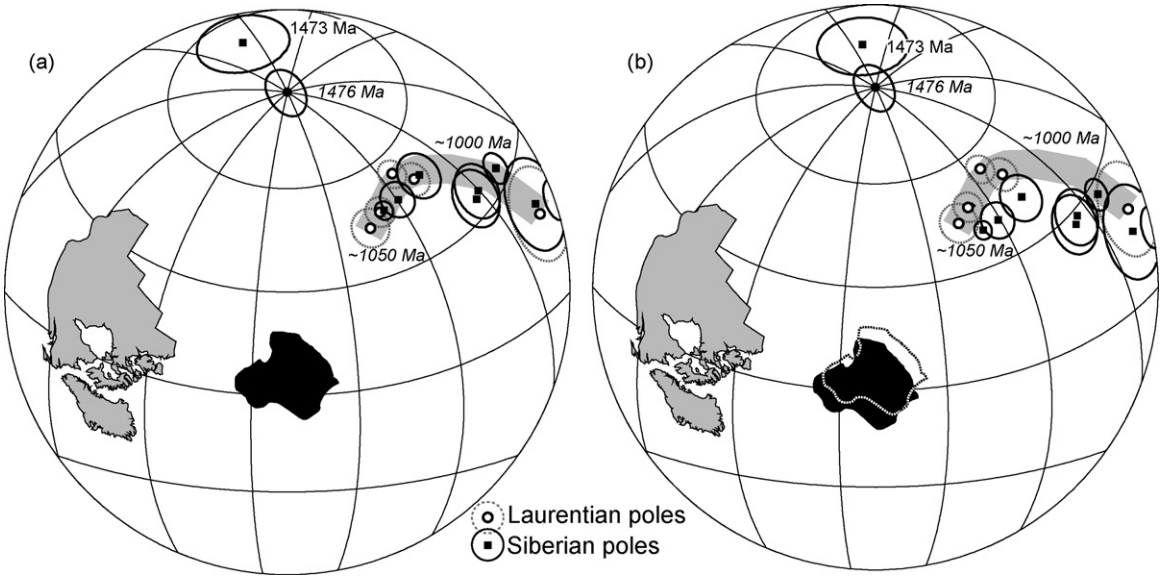


Fig. 8. Reconstructions of Laurentia and Siberia at ~1475 Ma. (a) Siberia is rotated to Laurentia about an Euler pole (+anticlockwise) at 65.0°N, 144.0°E, +141.8° according to the best fit between 1050 and 950 Ma APWPs (Pisarevsky and Natapov, 2003); (b) revised parameters to rotate Siberia to Laurentia to yield the best fit at both c. 1475 and 1050–950 Ma (66.6°N, 139.3°E, +134.8°); revised position of Siberia is shown with craton outline only. Laurentia is rotated to the absolute framework in both reconstructions about a pole at 0°, 129°E, +103.2°, according to the St. Francois pole of Meert and Stuckey (2002).

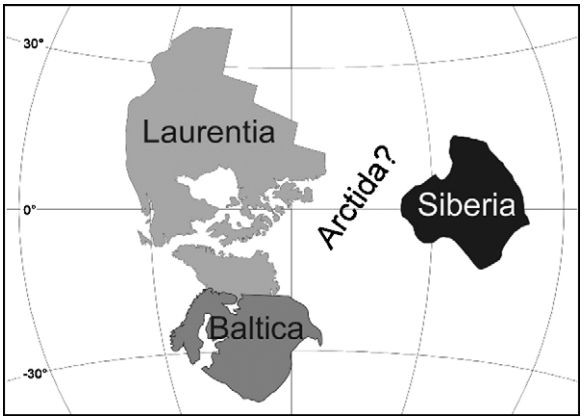


Fig. 9. Reconstruction of Laurentia, Siberia, and Baltica at ~1475 Ma. Euler rotation parameters for Laurentia and Siberia – see caption of Fig. 8b; Baltica to Laurentia: 36.1°, 1.7°, +43.15° (Salminen and Pesonen, 2007).

mine whether the Laurentia–Siberia reconstruction established for 1050–950 Ma (Pisarevsky and Natapov, 2003) is also applicable at ~1475 Ma. Fig. 8a shows that in this configuration the St. Francois and Kyutingde–Sololi poles are close to each other, but their ovals/circles of confidence do not overlap. However, a minor change to the rotation parameters for Siberia with respect to Laurentia results in a revised reconstruction that satisfies both the ~1475 Ma and 1050–950 Ma poles (Fig. 8b). We suggest therefore that Laurentia and Siberia were parts of the same supercontinent since at least ~1475 Ma until their separation in the mid-Neoproterozoic (Pisarevsky et al., 2008).

Ernst et al. (2000) reported a 1503 ± 5 Ma paleopole (Ku, Table 4 and Fig. 7) from the Kuonamka dykes of the eastern Anabar shield, Siberia, but its low precision and the absence of a Laurentian pole of similar age did not permit them to distinguish between various Laurentia–Siberia reconstructions. Veselovsky et al. (2006) reported a paleopole (Fo, Table 4, Fig. 7) from the Fomich River dykes and sills of the northern Anabar shield. Although it has a relatively imprecise Sm–Nd isochron age of 1513 ± 51 Ma, it also supports our reconstruction (Fig. 8).

Salminen and Pesonen (2007) obtained a new 1458 Ma pole for Baltica and conducted a similar test of the Laurentia–Baltica reconstruction for ~1265–1460 Ma. They concluded that the relative positions of these two continents did not change during this time interval. We used their Euler rotation parameters for Baltica with respect to Laurentia and reconstructed their position at 1475 Ma (Fig. 9). We argue therefore that Laurentia, Siberia, and Baltica were parts of the same supercontinent between 1475 Ma and the middle Neoproterozoic.

Acknowledgements

This research was supported in part by grants from the Russian Ministry of Education (MD 242.2007.5, NSH 3082.2008.5, RNP 2.2.1.1.7334), Russian Foundation for Basic Research (08-05-00245, 07-05-00339), and Russian Science Support Foundation. The authors are grateful for support of the Amakinskaya Expedition, ALROSA, during fieldwork in Siberia. Zircon and baddeleyite analyses were conducted using the SHRIMP ion microprobes at the John de Laeter Centre for Mass Spectrometry at Curtin University of Technology, in Perth, Australia. MW publishes with permission of the Executive Director of the Geological Survey of Western Australia. This paper is publication number 423 of the Tectonics Special Research Centre [number to be supplied after acceptance].

References

Bogdanova, S.V., Li, Z.X., Moores, E.M., Pisarevsky, S.A., 2008. Testing the Rodinia hypothesis: records in its building blocks. Precam. Res. 160, 1–4.
Buchan, K.L., Mertanen, S., Park, R.G., Pesonen, L.J., Elming, S.A., Abrahamsen, N., Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key paleomagnetic poles. Tectonophysics 319, 167–198.
Cawood, P.A., Kröner, A., Pisarevsky, S.A., 2006. Precambrian plate tectonics: criteria and evidence. GSA Today 16, 4–11.
Claoue-Long, J.C., Compston, W., Roberts, J., Fanning, C.M., 1995. Two Carboniferous ages: a comparison of SHRIMP zircon ages with conventional zircon ages and 40Ar/39Ar analysis. In: Berggren, W.A., Kent, D.V., Aubrey, M.P., Hardenbol, J. (Eds.), Geochronology, Time Scales, and Global Stratigraphic Correlation. SEPM Special Publication 54, pp. 3–21.
Compston, W., Williams, I.S., Meyer, C., 1984. U–Pb geochronology of zircons from lunar breccia 73217 using a sensitive high mass-resolution ion microprobe. J. Geophys. Res. Suppl. 89, B525–B534.
Condie, K.C., 1998. Episodic continental growth and supercontinents: a mantle avalanche connection? Earth Planet. Sci. Lett. 163, 97–108.
Dalziel, I.W.D., 1997. Neoproterozoic–Paleozoic geography and tectonics: review, hypothesis, environmental speculation. Geol. Soc. Am. Bull. 109, 16–42.
Ernst, R.E., Buchan, K.L., Hamilton, M.A., Okrugin, A.V., Tomshin, M.D., 2000. Integrated paleomagnetism and U–Pb geochronology of mafic dikes of the eastern Anabar Shield region, Siberia: implications for Mesoproterozoic paleolatitude of Siberia and comparison with Laurentia. J. Geol. 108, 381–401.
Evans, D.A.D., Pisarevsky, S.A., 2008. Plate tectonics on the early Earth? Weighing the paleomagnetic evidence. In: K. Condie, V. Pease (Eds.), When Did Plate Tectonics Begin? Geological Society of America Special Paper 440, 249–263, doi:10.1130/2008.2440(12).
Faure, G., 1986. Principles of Isotope Geology. John Wiley & Sons, New York, 589 pp.
Fisher, R.A., 1953. Dispersion on a sphere. Proceedings Royal Society London A217, 295–305.
Gallet, Y., Pavlov, V.E., Semikhatov, M.A., Petrov, P.Yu., 2000. Late Mesoproterozoic magnetostratigraphic results from Siberia: paleogeographic implications and magnetic field behaviour. J. Geophys. Res. 105, 16481–16499.
Gladkochub, D.P., Pisarevsky, S.A., Donskaya, T.V., Natapov, L.M., Mazukabzov, A.M., Stanevich, A.M., Slkyarov, E.V., 2006a. Siberian Craton and its evolution in terms of Rodinia hypothesis. Episodes 29, 169–174.
Gladkochub, D.P., Wingate, M.T.D., Pisarevsky, S.A., Donskaya, T.V., Mazukabzov, A.M., Ponomarchuk, V.A., Stanevich, A.M., 2006b. Mafic intrusions in southwestern Siberia and implications for a Neoproterozoic connection with Laurentia. Precam. Res. 147, 260–278.
Gorokhov, I.M., Semikhatov, M.A., Arakelyants, M.M., Fallick, E.A., Mel'nikov, N.N., Turchenko, T.L., Ivanovskaya, T.A., Zaitseva, T.S., Kutyavin, E.P., 2006. Rb–Sr, K–Ar, H- and O-isotope systematics of the Middle Riphean Shales from the Debengda Formation, the Olenek Uplift, North Siberia. Stratigr. Geol. Corr. 14, 260–274.
Gurevich, E.L., 1983. Paleomagnetic investigation of Precambrian deposits of the northern Siberian platform. In: Paleomagnetizm verkhnego dokembriya. Trudy VNIGRI, Leningrad, pp. 39–51.
Heaman, L.M., Lecheminant, A.N., 1993. Paragenesis and U–Pb systematics of baddeleyite (ZrO_2). Chem. Geol. 110, 95–126.
Iosifidi, A.G., Rodionov, V.P., 1986. Paleomagnetic Directions and Pole Positions: Data for the USSR – Issue 6. Soviet Geophysical Committee: World Data Center-B, Moscow.
Kats, A.G., Frolova, Z.B., 1986. New data on the stratigraphy of the Upper Proterozoic of the southern slope of the Olenek uplift. In: Pozdnii dokembrii i rannii paleozoi Sibiri. Sibirskaia platforma I vneshnyaya zona Altai-Sayanskoi skladchatoi oblasti. IgiG SO AN SSSR, Novosibirsk, pp. 65–84 (in Russian).
Khomentovsky, V.V., 2006. The Neoproterozoic of the northern Siberian craton. Russian Geol. Geophys. 47 (7), 862–877.
Krogh, T.E., Corfu, F., Davis, D.W., Dunning, G.R., Heaman, L.M., Kamo, S.L., Machado, N., Greenough, J.D., Nakamura, E., 1987. Precise U–Pb isotopic ages of diabase dykes and mafic to ultramafic rocks using trace amounts of baddeleyite and zircon. Geol. Assoc. Canada Spec. Paper 34, 147–152.
Krylov, A.J., Vishnevsky, A.N., Silin, Yu.I., Atrashenok, L.J., Avdzeiko, G.V., 1963. Absolute age of the rocks of the Anabar Shield. Geochimia 12, 1140–1144 (in Russian).
Li, Z.X., Bogdanova, S.V., Collins, A.S., Davidson, A., De Waele, B., Ernst, R.E., Fitzsimons, I.C.W., Fuck, R.A., Gladkochub, D.P., Jacobs, J., Karlstrom, K.E., Lu, S., Natapov, L.M., Pease, V., Pisarevsky, S.A., Thrane, K., Vernikovsky, V., 2008. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precamb. Res. 160, 179–210.
Ludwig, K.R., 2001. Squid 1.03. Berkeley Geochronology Centre Special Publication 2, 19 pp.
McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism. Geophys. J. Int. 103, 725–729.
Meert, J.G., 2002. Paleomagnetic evidence for a Paleo-Mesoproterozoic supercontinent Columbia. Gondwana Res. 5, 207–215.
Meert, J.G., Stuckey, W., 2002. Revisiting the paleomagnetism of the 1.476 Ga St. Francois Mountains igneous province, Missouri. Tectonics 21, 1007, doi:10.1029/2000TC001265.
Nelson, D.R., 1997. Compilation of SHRIMP U–Pb zircon geochronological data, 1996. W. Aust. Geol. Surv. Record 1997/2, 189p.
Pavlov, V.E., Gallet, Y., Petrov, P.Yu., Zhuravlev, D.Z., Shatsillo, A.V., 2002. Uy series and late Riphean sills of the Uchur-Maya area: isotopic and paleomagnetic data and the problem of the Rodinia supercontinent. Geotectonics 36, 278–292.

Pavlov, V.E., Gallet, Y., Shatsillo, A.V., 2000. Paleomagnetism of the upper Riphean Lakhandinskaya Group of the Uchuro-Maiskii area and the hypothesis of the Late Proterozoic supercontinent, Izvestiya. Phys. Solid Earth 36, 638–648.
Pesonen, L.J., Elming, S.-A., Mertanen, S., Pisarevsky, S.A., D'Agrella-Filho, M.S., Meert, J., Schmidt, P.W., Abrahamsen, N., Bylund, G., 2003. Paleomagnetic configuration of continents during the Proterozoic. Tectonophysics 375, 289–324.
Pidgeon, R.T., Furfaro, D., Kennedy, A.K., Nemchin, A.A., Van Bronswijk, W., 1984. Calibration of zircon standards for the Curtin SHRIMP II. In: Abstracts 8th International Conference on Geochronology, Cosmochronology, and Isotope Geology, Berkeley, USA, US Geol. Surv. Circ. 1107, p. 251.
Pisarevsky, S.A., 2005. New edition of the Global Paleomagnetic Database. EOS 86, 170.
Pisarevsky, S.A., Natapov, L.M., 2003. Siberia and Rodinia. Tectonophysics 375, 221–245.
Pisarevsky, S.A., Natapov, L.M., Donskaya, T.V., Gladkochub, D.P., Vernikovsky, V.A., 2008. Proterozoic Siberia: a promontory of Rodinia. Precam. Res. 160, 66–76.
Pisarevsky, S.A., Wingate, M.T.D., Powell, C. McA., Johnson, S., Evans, D.A.D., 2003. Models of Rodinia assembly and fragmentation. In: Yoshida, M., Windley, B., Dasgupta, S. (Eds.), Proterozoic East Gondwana: Supercontinent Assembly and Breakup. Geological Society of London Special Publication No 206, 35–55.
Ponomarchuk, V.A., Shenfil, V.Yu., Yakshin, M.S., Lebedev, Yu.N., Kuzmin, D.S., Morozova, I.P., 1994. The direct K–Ar dating of stromatolite of Olenek Uplift. Dokl. RAN 339, 378–381 (in Russian).
Rainbird, R.H., Stern, R.A., Khudoley, A.K., Kropachev, A.P., Heaman, L.M., Sukhorukov, V.I., 1998. U–Pb geochronology of Riphean sandstone and gabbro from southeast Siberia and its bearing on the Laurentia–Siberia connection. Earth Planet. Sci. Lett. 164, 409–420.
Rogers, J.J.W., 1996. A history of continents in the past three billion years. J. Geol. 104, 91–107.
Rosen, O.M., 2002. Siberian craton – a fragment of a Paleoproterozoic supercontinent. Russian J. Earth Sci. 4, 103–119.
Salminen, J., Pesonen, L.J., 2007. Paleomagnetic and rock magnetic study of the Mesoproterozoic sill, Valaam island, Russian Karelia. Precam. Res. 159, 212–230.
Sklyarov, E.V., Gladkochub, D.P., Mazukabzov, A.M., Menshagin, Y.V., Watanabe, T., Pisarevsky, S.A., 2003. Neoproterozoic mafic dike swarms of the Sharyzhalgai metamorphic massif (southern Siberian craton). Precamb. Res. 122, 359–376.
Stacey, J.S., Kramers, J.D., 1975. Approximation of terrestrial lead isotopic evolution by a two-stage model. Earth Planet. Sci. Lett. 26, 207–221.
Stanevich, A.M., Maksimova, E.N., Kornilova, T.A., Gladkochub, D.P., Mazukabzov, A.M., Donskaya, T.V., 2009. Microfossils of the Arymas and Debengde Formations of the Riphean of the Olenek Uplift: age and suggested nature. Stratigr. Geol. Corr. 17 (n. 1), 23–40.
Steiger, R.H., Jäger, E., 1977. Subcommision on geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth Planet. Sci. Lett. 36, 359–362.
Veselovsky, R.V., Petrov, P.Yu., Karpenko, S.F., Kostitsyn, Yu.A., Pavlov, V.E., 2006. New paleomagnetic and isotopic data on the Mesoproterozoic igneous complex on the northern slope of the Anabar Massif. Doklady Earth Sci. 411, 1190–1194.
Wingate, M.T.D., Campbell, I.H., Compston, W., Gibson, G.M., 1998. Ion microprobe U–Pb ages for Neoproterozoic basaltic magmatism in south-central Australia and implications for the breakup of Rodinia. Precam. Res. 87, 135–159.
Wingate, M.T.D., Compston, W., 2000. Crystal orientation effects during ion microprobe analysis of baddeleyite. Chem. Geol. 168, 75–97.
Zhao, G., Sun, M., Wilde, S.A., Li, S., 2004. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth-Sci. Rev. 67, 91–123.
Zonenshain, L.P., Kuzmin, M.I., Natapov, L.M., 1990. Geology of the USSR: a plate tectonic synthesis. Geodynamic Monograph, American Geophysical Union, Washington, Series 21, 242 pp.